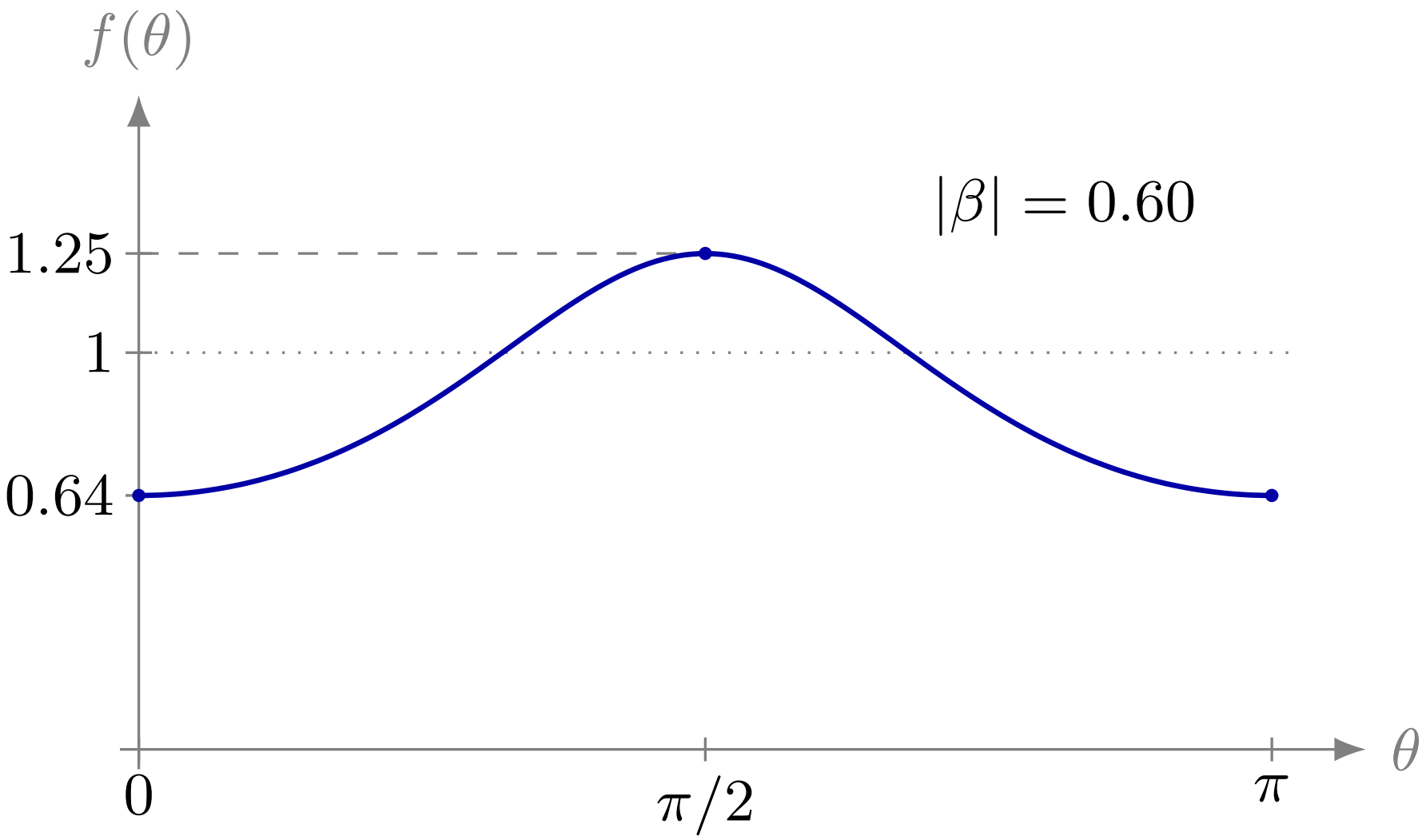

$$f = |\boldsymbol{E}'|/(k|q|/r'^2)$$